



THE UNIVERSITY OF
MELBOURNE

ACCT90002

Financial Statement

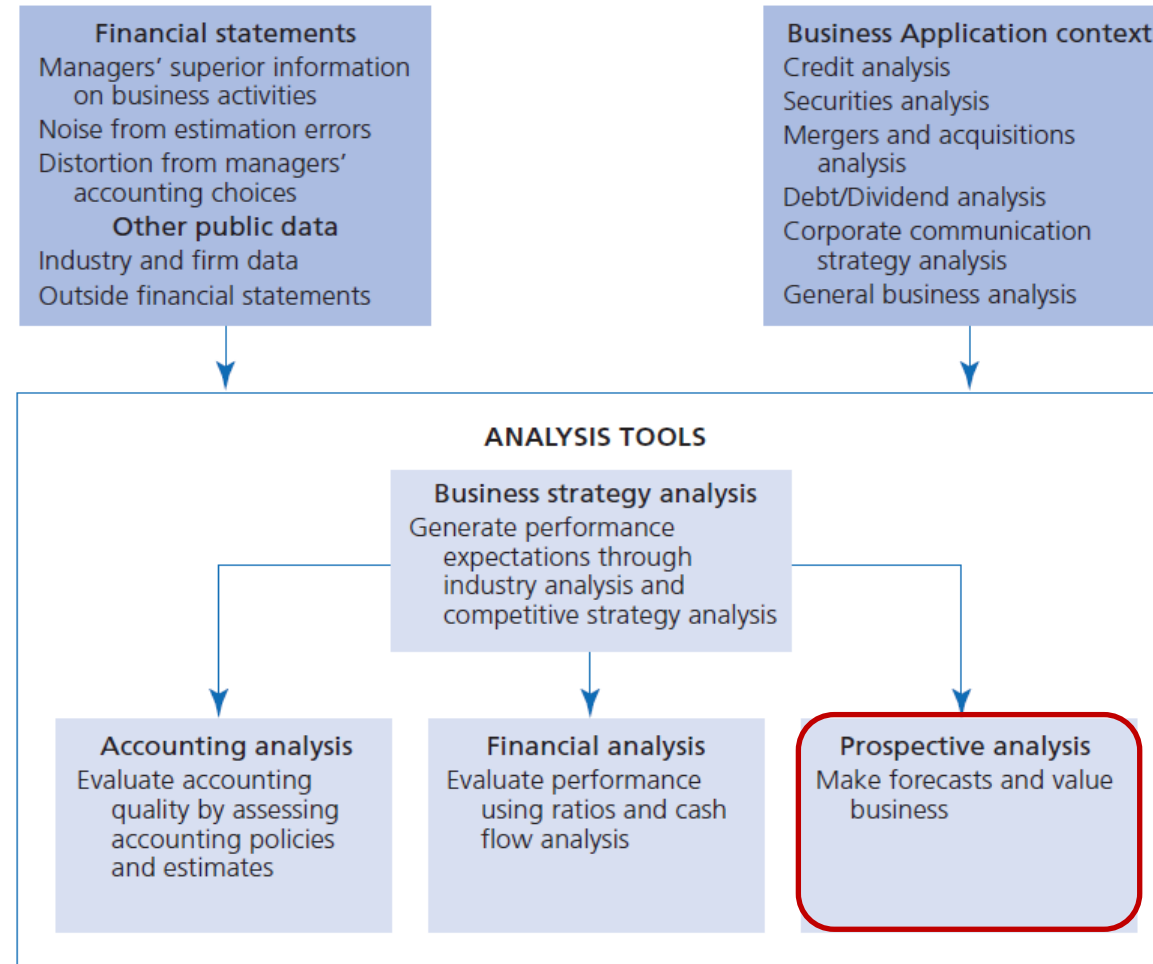
Analysis

Week 10: Valuation (2)



Framework - Financial Statement Analysis

FIGURE 1.3 Analysis using financial statements



Common fundamental valuation models

1. Discounted dividends:

- Value of equity = present value (PV) of forecasted **future dividends**

2. Discounted cash flow:

- Value of equity = PV of multiple year forecasts of **free cash flows to equity claim holders**

3. Discounted abnormal earnings (residual income):

- Value of equity = sum of **equity book value** and discounted forecasts of **abnormal earnings**

In principle – all three formulae give the same firm value

Discounted Abnormal Earnings (AE) Valuation

- Assuming 'clean surplus accounting' [all equity effects (**other than capital transactions, i.e., non-owner sources**) flow through the income statement]:

$$DIV_1 = NI_1 + BVE_0 - BVE_1$$

- DIV_1 = Expected dividends (**both** cash dividends and share issues/repurchases)
- NI_1 = Expected net income available to shareholders
- BVE_0 = Book value of equity at the **beginning** of the year
- BVE_1 = Expected book value of equity for **existing** shareholders at the **end** of the year

DDM to Discounted Abnormal Earnings (AE) Valuation model

APPENDIX: RECONCILING THE DISCOUNTED DIVIDENDS AND DISCOUNTED ABNORMAL EARNINGS MODELS

To derive the discounted abnormal earnings model from the dividend discount model, consider the following two-period valuation:

$$\text{Equity value} = \frac{DIV_1}{(1+r_e)} + \frac{DIV_2}{(1+r_e)^2}$$

With clean surplus accounting, dividends (DIV) can be expressed as a function of net income (NI) and the book value of equity (BVE):

$$DIV_t = NI_t + BVE_{t-1} - BVE_t$$

Substituting this expression into the dividend discount model yields the following:

$$\text{Equity value} = \frac{NI_1 + BVE_0 - BVE_1}{(1+r_e)} + \frac{NI_2 + BVE_1 - BVE_2}{(1+r_e)^2}$$

This can be rewritten as follows:

$$\begin{aligned} \text{Equity value} &= \frac{NI_1 - r_e BVE_0 + BVE_0(1+r_e) - BVE_1}{(1+r_e)} + \frac{NI_2 - r_e BVE_1 + BVE_1(1+r_e) - BVE_2}{(1+r_e)^2} \\ &= BVE_0 + \frac{NI_1 - r_e BVE_0}{(1+r_e)} + \frac{NI_2 - r_e BVE_1}{(1+r_e)^2} - \frac{BVE_2}{(1+r_e)^2} \end{aligned}$$

The value of equity is therefore the current book value plus the present value of future abnormal earnings. As the forecast horizon expands, the final term (the present value of liquidating book value) becomes inconsequential.

Discounted Abnormal Earnings (AE) Valuation

- Substitute above identity into the Dividend Discount Model (DDM) and equity value is re-written as follows:

$$Equity_value = BVE_0 + \frac{NI_1 - r_e \times BVE_0}{(1 + r_e)} + \frac{NI_2 - r_e \times BVE_1}{(1 + r_e)^2} + \dots$$

Fundamental value of equity = **Book value of equity** + **PV of future abnormal earnings**

- **Abnormal Earnings (AE)** = **net income** adjusted for (minus) a **capital charge** (**cost of equity capital** * opening book value of equity)

Discounted Abnormal Earnings (AE) Valuation

$$Equity_value = BVE_0 + \frac{NI_1 - r_e \times BVE_0}{(1 + r_e)} + \frac{NI_2 - r_e \times BVE_1}{(1 + r_e)^2} + \dots$$

- Deviation of a firm's **fundamental** value from its **book** value (i.e. **goodwill**) depends on its ability to generate “abnormal earnings”
- Research indicates that the model is **a better predictor** of value than the traditional multiples-based approaches mentioned earlier

AE model implementation

3-year horizon

$$\text{Equity_value} = \overset{(1)}{BVE_0} + \overset{(2)}{\frac{NI_1 - r_e \times BVE_0}{(1+r_e)}} + \overset{(3)}{\frac{NI_2 - r_e \times BVE_1}{(1+r_e)^2}} + \overset{(4)}{\frac{NI_3 - r_e \times BVE_2}{(1+r_e)^3}} + \overset{(5)}{TV}$$

- Start with the opening book value of equity (BVE_0)
(1)
- Calculate the **sum** of discounted abnormal earnings over a pre-defined horizon (here, 3 years) [(2)+(3)+(4)]
- Calculate the terminal value (TV) (5) ←

AE model implementation

terminal value

$$Equity_value = BVE_0 + \frac{NI_1 - r_e \times BVE_0}{(1 + r_e)} + \frac{NI_2 - r_e \times BVE_1}{(1 + r_e)^2} + \frac{NI_3 - r_e \times BVE_2}{(1 + r_e)^3} + TV$$

$$TV = \frac{(1 + g) * (NI_3 - r_e \times BVE_2)}{(r_e - g)} \div (1 + r_e)^3$$

➤ We have to project what happens to **Abnormal Earnings** (AE) after year 3 → assume an **infinite** yearly growth rate in AE.

Steps:

1. Project AE in year 4 (multiply AE year 3 by 1+g)
2. Discount back to year 3 with (r-g)
3. Do not forget to discount back the terminal value to year 0!

Abnormal earnings = residual income = Ohlson model

Ohlson, J. A. 1995. Earnings, book values, and dividends in equity valuation. *Contemporary Accounting Research* 11 (2): 661-687.

Note: The above terminology (abnormal earnings, residual income, Ohlson's model) is often used interchangeably

A practical problem

In practice, analysts must deal with a number of issues having an important effect on valuation, including:

- Negative book values. Start-up firms and firms in certain industrial sectors, among others, may have negative book equity (so what is capital charge?).

Solutions:

1. Value the firm's **assets** (abnormal NOPAT); subtract value of net debt to arrive at equity value estimate
2. Undo accountants' conservatism that *may* have led to negative book value

AE model: Valuing Assets

- Used to value **assets**:

$$Asset_Value = BVA_0 + \frac{NOPAT_1 - WACC \times BVA_0}{(1 + WACC)} + \frac{NOPAT_2 - WACC \times BVA_1}{(1 + WACC)^2} + \frac{NOPAT_3 - WACC \times BVA_2}{(1 + WACC)^3} + \dots$$

- BVA = book value of the firm's net (**operating**) assets
- NOPAT = net **operating** profit after tax
- WACC = weighted average cost of debt and equity capital

From the Asset Value above one can **deduct net debt** to generate an estimate of the **value of equity** [$A = L + E$]



Reformulating the balance sheet and income statement (variable definitions)

➤ **Net interest expense after tax (NIEAT):**

➤ $(\text{Interest Expense} - \text{Interest Income}) * (1 - \text{Tax Rate})$

➤ **Net operating profit after taxes (NOPAT):**

➤ $\text{Net income} + \text{Net interest expense after tax}$

➤ **Operating working capital (OWC):**

➤ $(\text{CA} - \text{Cash and marketable securities}) - (\text{CL} - \text{Short term debt})$

➤ **Net long term (non-current) assets (NLTA):**

➤ $\text{Total long term assets} - \text{non-interest bearing long term liabilities (e.g., tax liabilities, P-E benefit obligations, provisions, etc.)}$

➤ **Net debt:**

➤ $\text{Total interest bearing liabilities} - \text{cash and marketable securities}$

➤ **Net (operating) assets (BVA):** $\text{OWC} + \text{NLTA}$

➤ **Net capital:** $\text{Net debt} + \text{shareholders' equity}$

Implementation: discount rate

- When valuing **equity**, we have to discount the abnormal earnings available to equity holders
→ use **“cost of equity capital”**
- When valuing **assets**, we have to discount the abnormal earnings (NOPAT) available to equity *and* debt holders
→ use **“weighted average cost of capital (WACC)”**

Cost of equity capital

CAPM:

$$r_e = r_f + \beta [E(r_m) - r_f]$$

Risk-free rate

Premium for “**systematic**” risk

Estimating Weighted Average Cost of Capital (WACC)

➤ The appropriate discount rate to value a company's **assets** (debt plus equity) is the WACC, which takes into account debt and equity sources of financing:

$$\text{WACC} = \frac{\text{Net debt}}{\text{Net debt} + \text{Equity}} (1 - \text{Tax rate}) \times \text{Cost of debt} + \frac{\text{Equity}}{\text{Net debt} + \text{Equity}} \text{Cost of equity}$$

➤ Cost of debt (with a consideration of tax) = (net) interest rate on (net) debt
= NIEAT/ND:

Two Abnormal Earnings models

Valuing equity using the Abnormal Earnings model:

$$Equity_value = BVE_0 + \frac{NI_1 - r_e \times BVE_0}{(1 + r_e)} + \frac{NI_2 - r_e \times BVE_1}{(1 + r_e)^2} + \dots$$

Valuing assets using the Abnormal NOPAT model:

$$Asset_Value = BVA_0 + \frac{NOPAT_1 - WACC \times BVA_0}{(1 + WACC)} + \frac{NOPAT_2 - WACC \times BVA_1}{(1 + WACC)^2} + \dots$$

Terminal value

- The **terminal value** is the final year of the forecast and represents the PV of future abnormal earnings or free cash flows **for the remainder of the firm's life**
- Forecasting issues:
 - Assumptions about sales growth beyond the terminal year may be irrelevant because of **competitive equilibrium** [**TV = 0!**]
 - Terminal values may be forecasted with growth in abnormal earnings or cash flows **at a constant rate**
 - Selecting the terminal year: five- to ten-year forecast horizon should suffice for most firms

Terminal value: remember from earlier ...

$$TV = \frac{(1+g) * (NI_3 - r_e \times BVE_2)}{(r_e - g)} \div (1+r_e)^3$$

- We have to project what happens to AE after year 3 → assume an infinite yearly growth rate in AE.
e.g., $g=0\%$, $g=4\%$ (inflation)
- First project AE in year 4 (multiply AE year 3 by $1+g$)
- Discount back to year 3 with $(r-g)$
- Do not forget to discount back the terminal value to year 0!

RELATIVE VALUATION METHODS



P/E multiple

$$\text{P/E multiple} = \frac{\text{Share price}}{\text{EPS}}$$

Earnings (*an important fundamental*) reflecting:

- Industry growth,
- A firm's competitive position in the industry,
- Quality of management,
- New value generated (support dividend and stock repurchases)

A firm's economic value coheres with its fundamentals



Valuation using price multiples: A numerical example

Suppose:

Company ABC current **stock price** is \$100 and **EPS** of \$5

→ P/E multiple = **20** (*stock is currently valued at 20 times its earnings*)

Forecasted EPS for next year will be \$7

Then forecasted stock price will be \$140 ($= \$7 * 20$)

Commonly used multiples: **P/E multiple** & **P/B multiple** (price-to-book equity ratio)

Company Overview

- **Inditex:** Spanish multinational, parent of **Zara** and other fashion brands.
- **Fast Retailing (Uniqlo):** A leading global apparel retailer headquartered in Japan, best known as the parent company of **Uniqlo**.
- **H&M:** Swedish multinational clothing retailer known for fast fashion.



Metric	Inditex (EUR)	Fast Retailing (JPY)	H&M (SEK)
Financial year end	Jan 31, 2025	Aug 31, 2024	Nov 30, 2024
Net income (millions)	5900	393605	11621
Shares outstanding (millions)	3116	307	1612
Earnings per share	1.88	1212.88	7.21
Share price	47.81	36000	160
Price-to-earnings multiple	25.4	29.7	22.2
Beginning common equity (millions)	18740	1200000	66000
Return on beginning book equity (ROE)	31.50%	32.80%	17.60%
Ending common equity (millions)	19700	1300000	70000
Common equity growth	5.10%	8.30%	6.10%
Ending common equity per share	6.32	4234.53	43.43
Price-to-ending common equity multiple	7.56	8.5	3.68
Market capitalization (millions)	149120	11052000	257920



Analyze P/E multiple

Q: Why do firms in the same industry have such different P/E multiples?

- Investors willing to pay more for **Fast Retailing's** future earnings than for **Inditex's**?
- **H&M's** earnings perceived to be less valuable than **Inditex's**?

A:

- ❖ Three firms will generate very different earnings (and hence potential dividend payments) in the future.
- ❖ Investors perceive differences in riskiness of their future earnings.



Earning growth

Fast Retailing > Inditex > H&M

Company	2019	2020	2021	2022	2023	2024	5-Year CAGR
Inditex	€ 1.00	€ 0.35	€ 0.79	€ 1.05	€ 1.30	€ 1.88	13.46%
Fast Retailing	¥554.37	¥295.1	¥554.4	¥891.8	¥966.1	¥1212.9	16.95%
H&M	SEK 6.81	SEK 2.16	SEK 5.37	SEK 7.21	SEK 6.81	SEK 7.21	1.15%



Volatility/risk

Stock Price Volatility (5-Year Beta)

Company	5-Year Beta	Interpretation
Inditex	1.06	Slightly above market volatility
Fast Retailing	0.45	Lower than market volatility
H&M	1.20	Higher than market volatility

Risk of Fast Retailing < Inditex < H&M

Volatility/risk of the companies

They have different financial policies (i.e., financial leverage)

Fiscal Year	Inditex	Fast Retailing	H&M
2020	52.0%	53.7%	68.7%
2021	50.5%	49.3%	66.6%
2022	49.8%	47.2%	72.1%
2023	48.2%	46.5%	73.7%
2024	47.0%	44.5%	74.5%

- **Inditex:** Demonstrates a consistent decrease in leverage over the years, indicating a strong balance sheet with prudent financial management.
- **Fast Retailing:** Shows a steady decline in leverage, reflecting effective debt management and a solid financial position.
- **H&M:** Exhibits higher leverage ratios compared to its peers, suggesting a greater reliance on debt financing. This could be indicative of strategic investments or operational challenges.



Be cautious when applying P/E multiple

Problem:

P/E multiples are highly sensitive to the current earnings number

Solution:

Focusing on continuing operations

Using either ***earnings from continuing operations*** or ***earnings excluding one-time gains or losses*** as the denominator in P/E computations

PB multiple (Price-to-book equity ratio)

$$\text{Price-to-book equity ratio} = \frac{\text{Share price}}{\text{Book value of common equity per share}}$$

Book equity representing:

- Stock of capital has been invested and reinvested in the firm
- Providing the base on which firms generate earnings for SH
- All else being equal, firms with higher capital bases should generate higher future earnings and hence higher future dividends

Analyze price-to-book equity ratio

Metric	Inditex (EUR)	Fast Retailing (JPY)	H&M (SEK)
Price-to-ending common equity multiple	7.56	8.5	3.68
Return on beginning book equity (ROE)	31.50%	32.80%	17.60%

Explanations:

- Fast Retailing is expected to have a higher ROE (generating higher future earnings from its book equity than others)
In 2024 ROE, Fast Retailing > Inditex > H&M
- Differences across them in the riskiness of future earnings also matter

Valuation models: price multiples

Widely used as it is simple to execute (*simplicity is beauty*):

1. Select a measure of performance or value:
such as earnings, sales, cash flows, book equity, book assets
2. Estimate price multiples for **comparable** firms using the measure of performance or value
3. Apply the **comparable firm multiple** to the performance or value measure of the firm being analysed

Comparative valuation



Valuation using price multiples: An example (Simple average)

- To find the value of JB Hi-Fi (ASX: JBH) implied by other listed companies, we first find its comparable companies.
- We use other retail companies as its comp set: Super Retail and Harvey Norman
- We then collect the earnings and price data on the comparable companies and calculate the P/E for all comparable companies (from *Yahoo Finance*)

	Price (22/08/2023)	Earnings (22/23E)	P/E
SUL	\$13.13	\$1.17	11.22x
HVN	\$3.79	\$0.65	5.83x

- We decide to use average P/E (8.53x) and apply average P/E to JB Hi-Fi's '22 earnings (\$4.78) to value its stock.
- JB Hi-Fi's value is \$40.77 (8.53x\$4.78). It appears that it is slightly overvalued (its market price was \$46.67 on 22/08/23)



Another example—Fast Retailing (Weighted average)

We want to value Fast Retailing and choose Inditex and H&M as comparable stocks

	Market capitalization (USD billions)	Weightin g	PE	Weighted PE	PB	Weighted PB
Inditex	164	87.00%	25.4	22.1	7.56	6.58
H&M	24.5	13.00%	22.2	2.9	3.68	0.48
Total	188.5	100.00%		25		7.06

Fast Retailing

EPS = ¥1,212.88

Benchmark PE = 25

Price = 1,212.88 * 25 = ¥30,322

BVS = ¥4,234.35

Benchmark PB = 7.06

Price = 4,234.35 * 7.06 = ¥ 29,894.51

Valuation using multiples: Precautions

- Selecting comparable firms
 - Difficult to identify comparable firms even within an industry due to different strategies, growth opportunities, and profitability
 - Industry averages may be used instead

Valuation using multiples: Precautions

- Firms with poor performance
 - ❑ Negative price multiples are meaningless
 - ❑ Adjustments for transitory shocks (i.e., write-off and special item) and use **operating earnings**
- Adjustments for leverage
 - ❑ Issues arise from lack of consistency between numerator and denominator
 - ❑ e.g. **price** to **sales**, **price** to **operating earnings** (i.e., performance before servicing debt)



P/B multiple revisited (assume future abnormal earnings perpetuity)

$$\frac{\text{Equity value}}{\text{BVE}} = 1 + \frac{E(\text{ROE}) - r_e}{r_e - g_{\text{BVE}}}$$

where:

- $E(\text{ROE})$ = the firm's steady state expected return on beginning book common equity
- r_e is the required shareholder return
- g_{BVE} is the expected steady state growth in book equity, driven by firm size and a firm's competitive advantage (converging over a long term, about 3-4% per year for US firms)

P/B multiple revisited

$$\frac{\text{Equity value}}{\text{BVE}} = 1 + \frac{E(\text{ROE}) - r_e}{r_e - g_{\text{BVE}}}$$

What if...

$$E(\text{ROE}) = r_e$$

$$E(\text{ROE}) > r_e$$

$$E(\text{ROE}) < r_e$$

P/B multiple revisited

If $E(\text{ROE}) = r_e$

EXHIBIT 13 Hypothetical value-to-book multiple for firm with expected ROE of 10%, 10% return required by investors, and varying growth in book equity

Growth Rate	Value-to-Book Multiple for ROE of 10%
1%	1.00
3%	1.00
5%	1.00
7%	1.00
9%	1.00
11%	1.00

$E(\text{ROE}) = 15\%; r_e = 10\%$

EXHIBIT 12 Hypothetical value-to-book multiple for firm with expected ROE of 15%, 10% return required by investors, and varying growth in book equity

Growth Rate	Value-to-Book Multiple for ROE of 15%
1%	1.56
3%	1.71
5%	2.00
7%	2.67
9%	6.00
11%	-4.00

The perpetuity valuation makes sense only when assumed growth rate is less than the discount rate (a firm cannot grow infinitely at a high growth rate or it will eventually take over the world)

P/B multiple revisited

$$E(\text{ROE}) = 8\%; r_e = 10\%$$

EXHIBIT 14 Hypothetical value-to book multiple for firm with expected ROE of 8%, 10% return required by investors, and varying growth in book equity

Growth Rate	Value-to-Book Multiple for ROE of 8%
1%	0.78
3%	0.71
5%	0.60
7%	0.33

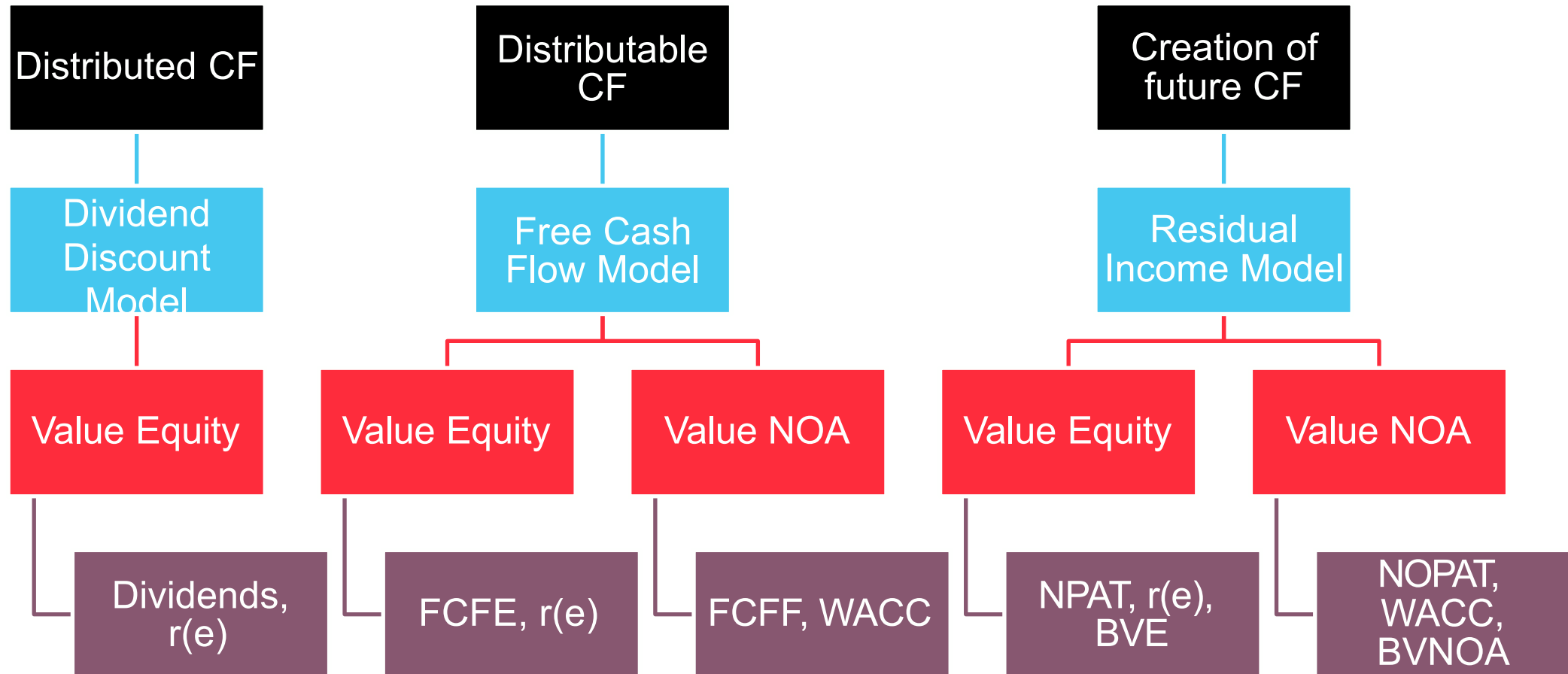
*Growth actually destroys value and reduces the firm's value-to-book multiple;
This is analogous to selling a product at a loss:
The more you sell, the greater the loss*

P/B multiple revisited

In summary:

- Growth is **only valuable** if a firm is expected to generate future ROEs greater than investors' required return
- If a firm's future ROE is expected to be lower than the expected return, growth **destroys** shareholder value, leading the firm to have a PB less than 1

Summing Up





Conclusion

- **Final comments and takeaways**
 - ✓ Analysts have a wide variety of choices when it comes to the valuation of securities
 - ✓ It is crucial to understand the pros and cons of any chosen strategy to value a security, and to understand exactly **what** is being valued
 - ✓ It is crucial to find a comparable company (a group of comparable companies) when using multiple approach to do valuation.



THE UNIVERSITY OF
MELBOURNE

See you at the seminar

